ALGORITHM AND REPRESENTATIONS

1. What is an algorithm?

Algorithms can be understood as the correct and detailed ways to solve a problem. In Coding and Data Science, Algorithms are a set of finite, well-defined steps or instructions designed to solve or perform a computation.

> **Flow:** Input $\to$ Process (using data based on algorithms) $\to$ Output

What makes a good algorithm? (Requirements)

1. **Determinacy (Tính xác định):** Every step needs to be accurate and unambiguous.
2. **Finiteness (Tính hữu hạn):** The algorithm has to terminate after a limited number of steps.
3. **Efficiency (Tính hiệu quả):** Use resources (time, memory) in a reasonable and effective way.
4. **Generality (Tính tổng quát):** Can be applied to different cases/inputs, not just one.

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2. How to write Pseudo-code

Definition

Pseudo-code is a technique used to describe distinct steps of an algorithm in plain language that helps everyone understand the logic without worrying about specific programming syntax.

Main Constructs

- **SEQUENCE (Tuần tự):** Execute linearly from the start to the end.
- **WHILE (Vòng lặp "Trong khi"):** Check the condition at the **beginning** of a loop.
- **REPEAT - UNTIL:** Check the condition at the **bottom** of a loop.
- **FOR:** Another way of looping (usually with a counter).
- **IF-THEN-ELSE:** Two-way decision making.
- **CASE:** A generalization of IF-THEN-ELSE, used when there are multiple potential paths.

Keywords

Use standard capitalized keywords: `START`, `END`, `INPUT`, `PRINT`, `IF...THEN...ELSE`, `WHILE`.

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3. How to draw a Flowchart

Components of Flowchart

Shape	English Name	Vietnamese Name	Function
▨ (Oval)	**Oval / Terminal**	Hình Elip	Start or End of a program.
▱ (Slanted)	**Parallelogram**	Hình Bình Hành	Input or Output.
▭ (Rect)	**Rectangle**	Hình Chữ Nhật	Process, computation.

Shape	English Name	Vietnamese Name	Function
◇ (Diamond)	**Diamond**	Hình Thoi (Kim cương)	Condition or branching (Yes/No).
➡️	**Arrow**	Mũi tên	Flowline (Direction).

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4. PRACTICE: Quadratic Equation Solver

Problem: Solve $ax^2 + bx + c = 0$

Decomposition (Phân rã bài toán)

1. **Step 1:** Input variables `a`, `b`, `c`.
2. **Step 2:** Check condition if `a` is 0. If it is, it becomes a linear equation $bx+c=0$.
3. **Step 3:** If `a` is not 0, calculate Delta ($d = b^2 - 4ac$).
4. **Step 4:** Determine result based on Delta:
   - If $d < 0$: No real roots.
   - If $d = 0$: Same roots (double root) $x = -b/(2a)$.
   - If $d > 0$: Two distinct roots $x = \frac{-b \pm \sqrt{d}}{2a}$.

Pseudo-code (Final Version)

```
BEGIN
    INPUT a, b, c

    IF a = 0 THEN
        // Case: Linear equation bx + c = 0
        IF b = 0 THEN
            IF c = 0 THEN
                PRINT "Many roots"
            ELSE
                PRINT "No roots"
            ENDIF
        ELSE
            x = -c / b
            PRINT x
        ENDIF
    ELSE
        // Case: Quadratic equation
        d = b*b - 4*a*c

        IF d < 0 THEN
            PRINT "No real roots"
        ELSE IF d = 0 THEN
            x = -b / (2 * a)
            PRINT x
        ELSE
            x1 = (-b + sqrt(d)) / (2 * a)
            x2 = (-b - sqrt(d)) / (2 * a)
            PRINT "x1 = ", x1
            PRINT "x2 = ", x2
```

```
        ENDIF
    ENDIF
```